

Predicting Joke Preferences with an Ising Model on the Jester Dataset

CS179: Graphical Models, Course Project

1. Introduction

We study whether a pairwise Markov random field, specifically an Ising model, can predict which jokes a user will find funny on the Jester joke-rating dataset. Each joke is treated as a binary spin variable (funny or not funny for a given user), with per-joke bias terms h that capture how broadly liked a joke is, and a symmetric coupling matrix J that captures how pairs of jokes tend to be rated together. Treating the problem this way lets us ask a clean question: does modeling jokes as an interacting system beat scoring each joke on its own? We frame prediction as matrix completion. We hide one rating per user, fit the model on the rest, and ask it to recover the held-out rating. This gives us a direct, quantitative way to measure performance and to study how that performance responds to changes in the amount of data and in the model itself.

2. Resources and Implementation

Data. We use the Jester dataset of continuous joke ratings in the range -10 to 10. We keep the 50 most-rated jokes and the users who rated at least 80% of them, then binarize each rating as funny if positive and not funny otherwise. The result is a dense 1000 user by 50 joke matrix, which is well suited to a fully connected Ising model.

Tools. The model is implemented in Pyro (pyro-ppl) on top of PyTorch. We learn the parameters by stochastic variational inference using an AutoNormal guide and the Adam optimizer, with a Bernoulli likelihood whose logits come from the Ising effective field. Baselines, metrics, clustering, and plots use scikit-learn, SciPy, and Matplotlib. The variational training loop and the Ising effective-field formulation are our own implementation; the standard inference machinery (SVI, AutoNormal, Adam) and the metric and clustering routines are library functions. All experiments are reproducible from a single notebook with a fixed seed, and a requirements file pins the dependencies.

3. Evaluation

Protocol and metrics. We hold out exactly one rated joke per user as the test target and train on everything else. The same held-out split is scored for every model and baseline, so comparisons are fair. Because the data is imbalanced (73.4% of held-out ratings are positive), accuracy alone is misleading, so we report five metrics: accuracy, AUC, log loss, Brier score, and expected calibration error (ECE). AUC and log loss in particular reward a model for ranking and probability quality rather than for simply guessing the majority class.

Baselines. We compare against three reference strategies that capture the easy signal without any pairwise modeling: a global majority predictor, a per-joke majority predictor, and a user-mean predictor based on each user's own positive rate. The honest bar to beat is the user-mean baseline, which already reaches 79.8% accuracy simply because most ratings are positive and users differ in how generous they are.

Model	Acc	AUC	LogLoss	Brier	ECE
Global majority	74.9%	0.500	0.564	0.188	0.015
Per-joke majority	76.8%	0.666	0.525	0.172	0.024
User mean	79.8%	0.766	0.468	0.149	0.032
Ising model	80.9%	0.830	0.422	0.133	0.024

Table 1. Held-out performance. The Ising model leads on every metric. Note the gap pattern: the accuracy edge over the user-mean baseline is small (1.1 points), but the AUC edge is large (0.830 vs 0.766).

The Ising model versus the baselines. The Ising model wins on all five metrics, but the interesting part is *how* it wins. Its accuracy advantage over the user-mean baseline is only about one point, which on its own might suggest the pairwise structure barely helps. The probability-aware metrics tell a different and more favorable story: AUC rises from 0.766 to 0.830 and log loss falls from 0.468 to 0.422. In other words, the coupling terms mostly improve how well the model *ranks and scores* jokes by confidence, even when the hard 0.5 threshold decision changes for only a few held-out items. For a recommendation setting, where we care about ordering jokes by how likely a user is to enjoy them, AUC is the more relevant measure, and that is exactly where the graphical model earns its keep.

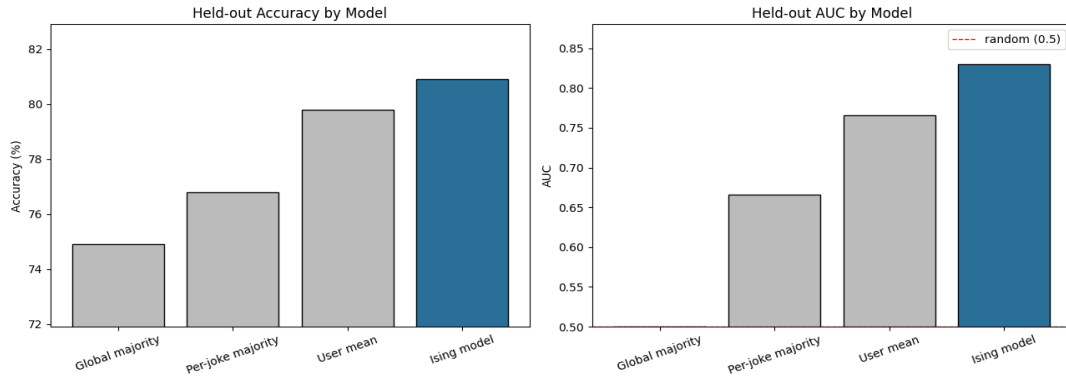


Figure 1. Accuracy (left) and AUC (right) across models. The accuracy bars are close together, while the AUC gap between the Ising model and the baselines is much wider.

Effect of training data. We next vary how much of the training data the model sees, from 10% up to 100%, repeating each setting over three random seeds and reporting the mean with error bars so the trend is not an artifact of a single lucky split. Performance climbs steadily and then begins to flatten. AUC in particular jumps from 0.677 at 10% of the data to 0.833 at full data, and the small error bars show the trend is stable. At the lowest data level the model sits near the per-joke baseline, then pulls clearly above it as more ratings become available, which is direct evidence that the pairwise couplings need enough co-rating observations before they can be estimated reliably.

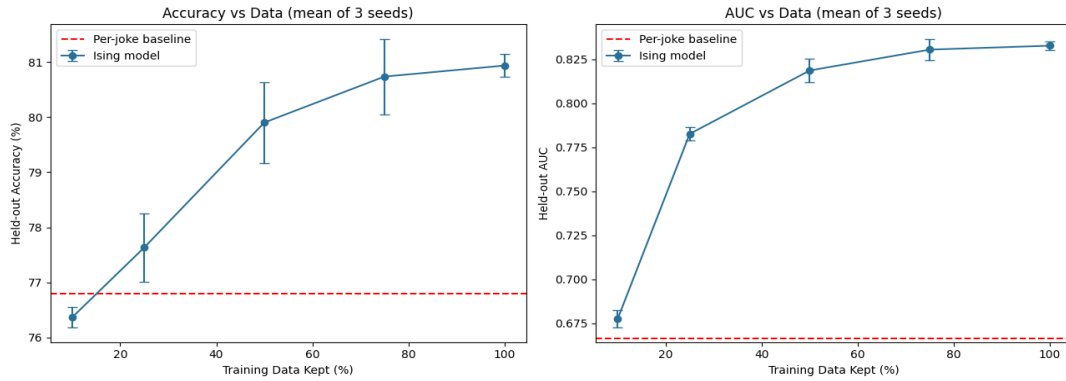


Figure 2. Accuracy (left) and AUC (right) versus training-data fraction, averaged over three seeds. The dashed line is the per-joke baseline. Both metrics rise then plateau; the AUC gain is the steeper and cleaner of the two.

Effect of changing the model. Finally we vary the model itself rather than the data, by sweeping the prior scale on the coupling matrix J . A very small scale pulls J toward zero, making jokes nearly independent and collapsing the model toward the per-joke baseline; a large scale allows strong pairwise interactions. Accuracy is almost flat across this sweep (roughly 80.9 to 81.0% for any non-trivial coupling), but AUC and log loss reveal a clear sweet spot at a scale of about 0.05, where AUC peaks at 0.838 and log loss bottoms out at 0.412. Below that the couplings are too constrained to capture joint structure, and above it the model begins to overfit noisy pairwise terms, so the probability quality degrades even though accuracy does not move. This confirms that the interaction terms, the whole reason for using a graphical model here, carry real predictive value when their strength is set sensibly.

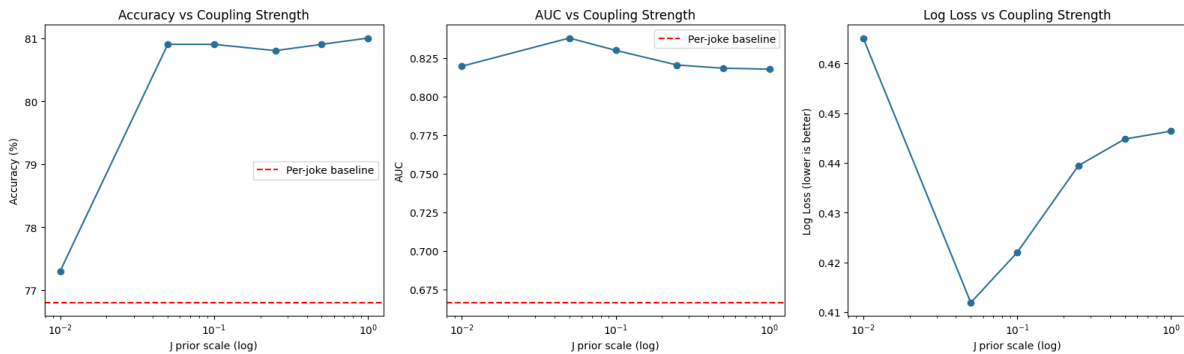


Figure 3. Accuracy, AUC, and log loss versus the coupling prior scale (log x-axis). Accuracy is flat, but AUC and log loss both identify a sweet spot near 0.05.

Calibration and structure. Beyond point performance, the model is reasonably well calibrated: binning held-out predictions by confidence and comparing to the observed positive rate gives an ECE of 0.024, with the calibration curve tracking the diagonal closely and most predictions landing in the high-confidence bins. Clustering the learned J matrix with hierarchical clustering also surfaces small groups of jokes that are rated similarly across users, showing that the couplings encode interpretable taste structure rather than noise. These two views are shown in Figure 4.

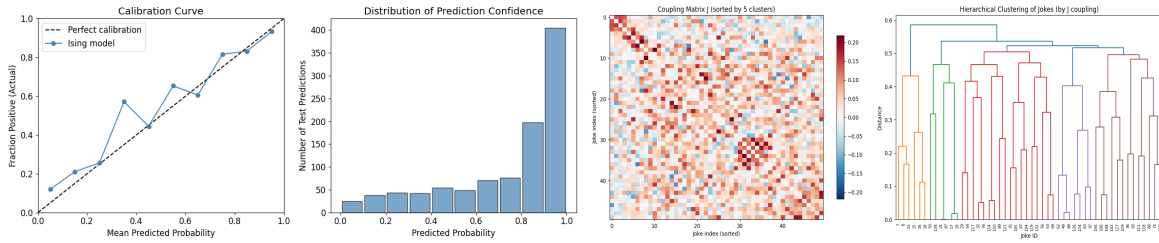


Figure 4. Left: calibration curve and confidence histogram (ECE = 0.024). Right: hierarchical clustering of jokes from the learned coupling matrix J.

4. Conclusion

An Ising model predicts held-out joke preferences on Jester at 80.9% accuracy and 0.830 AUC, beating global-majority, per-joke, and user-mean baselines on every metric we measured. The most useful finding is where the advantage lives: the pairwise couplings deliver a large gain in ranking and probability quality (AUC and log loss) while moving threshold accuracy only slightly, because much of the raw accuracy is already explained by overall joke popularity and per-user generosity. Performance improves with more training data and then plateaus, and varying the coupling strength exposes a clear sweet spot that confirms the interaction terms are doing real, tunable work. A natural next step would be to replace the variational point estimate with full posterior sampling to obtain uncertainty over J, and to compare against a low-rank matrix-factorization model on the same held-out split.